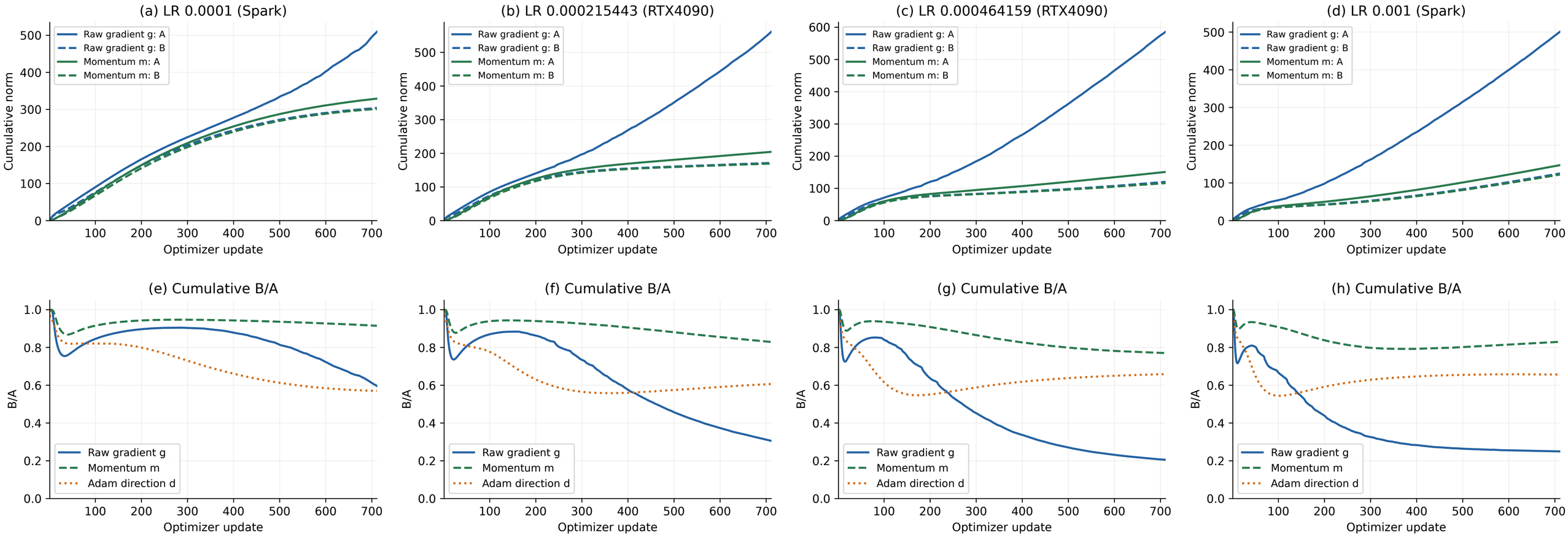
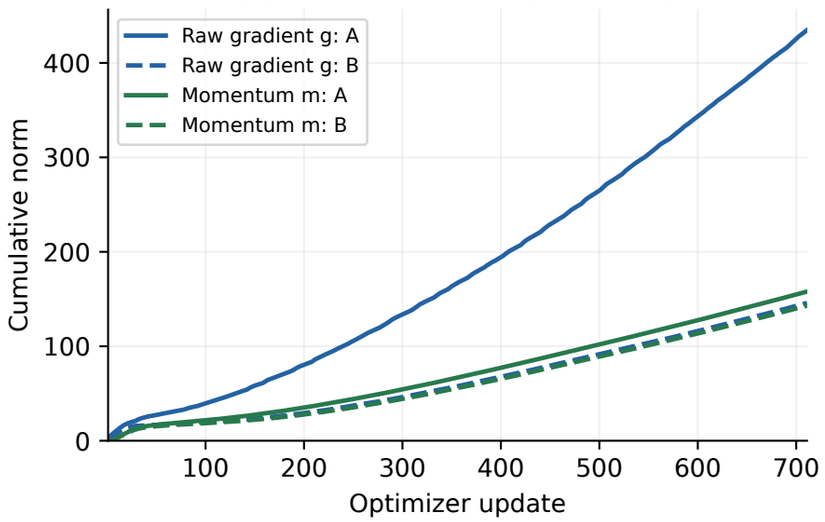
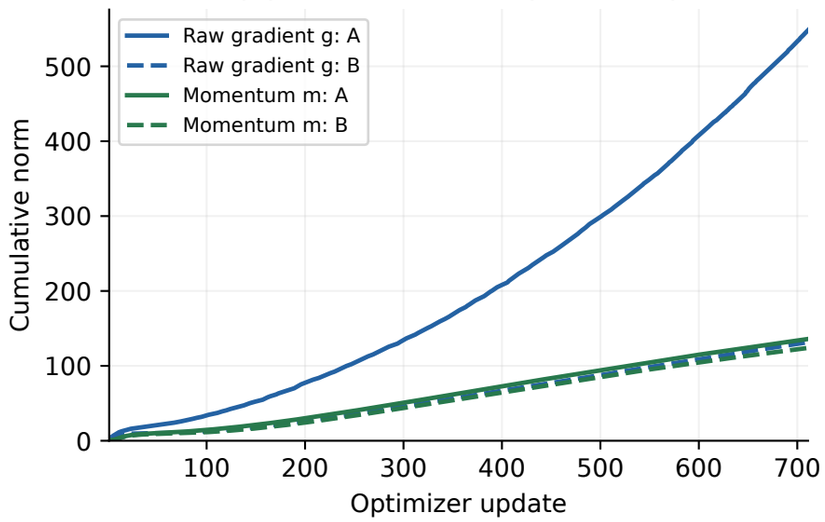
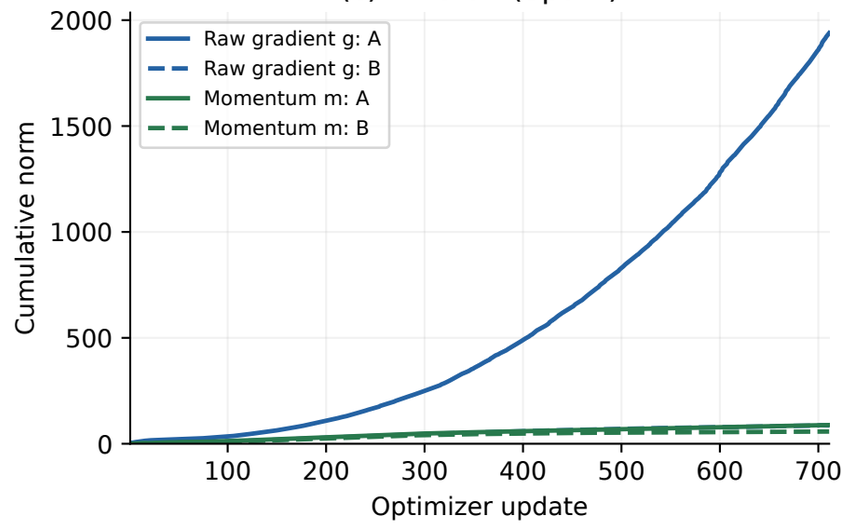
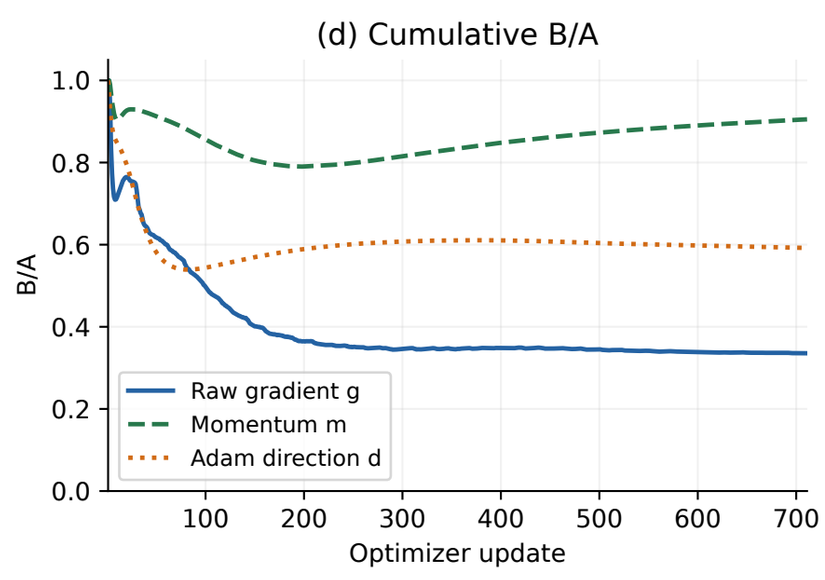

Pythia-14M: geometry within each LR (page 1/2)
(a) LR 0.0001 (Spark)
(b) LR 0.000215443 (RTX4090)
(c) LR 0.000464159 (RTX4090)
(d) LR 0.001 (Spark)
Raw gradient g: A
Raw gradient g: B
Momentum m: A
Momentum m: B
Cumulative norm
Optimizer update
(e) Cumulative B/A
(f) Cumulative B/A
(g) Cumulative B/A
(h) Cumulative B/A
Raw gradient g
Momentum m
Adam direction d
B/A
A = sum of norms; B = norm of sum. Top: raw gradient g and actual stored momentum m; solid A, dashed B.
Each top panel has its own LINEAR y-scale to emphasize shape. Lower row: g, m and adaptive d; common range.
All curves unsmoothed; updates 1-711 shown; update 712 retained in results. m uses clipped gradients; d excludes LR/decay.
Frozen initialization/order and seed 1234; hardware differs across original/new runs; no convergence or stability inference.

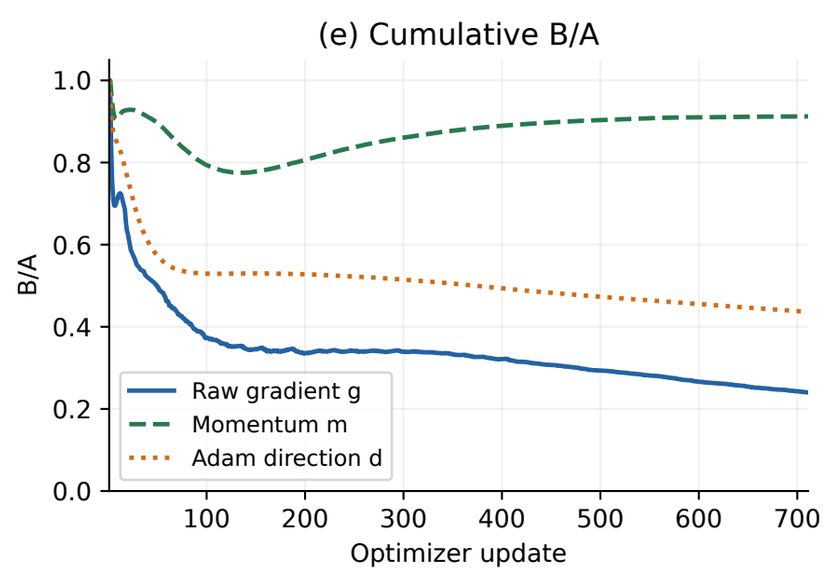
Pythia-14M: geometry within each LR (page 2/2)

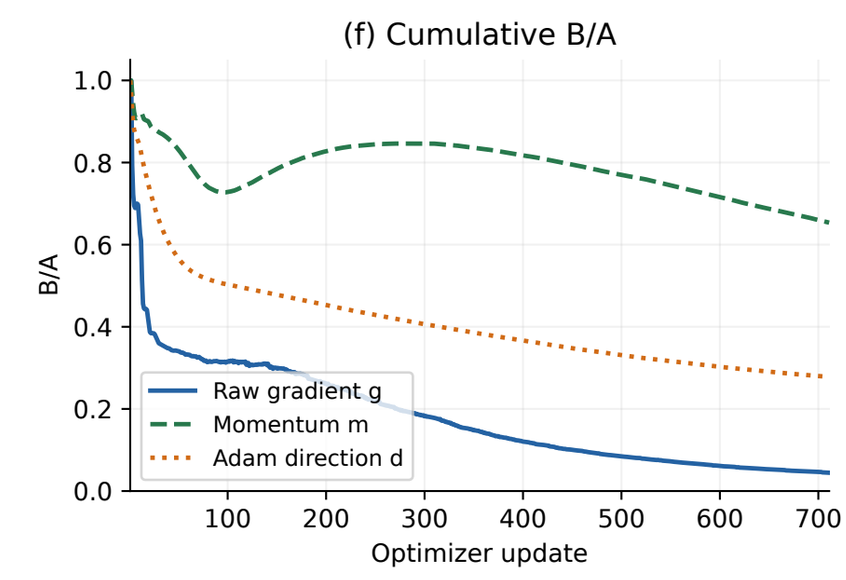












A = sum of norms; B = norm of sum. Top: raw gradient g and actual stored momentum m; solid A, dashed B.
Each top panel has its own LINEAR y-scale to emphasize shape. Lower row: g, m and adaptive d; common range.
All curves unsmoothed; updates 1-711 shown; update 712 retained in results. m uses clipped gradients; d excludes LR/decay.
Frozen initialization/order and seed 1234; hardware differs across original/new runs; no convergence or stability inference.